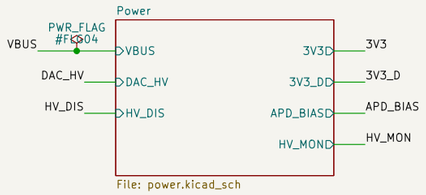
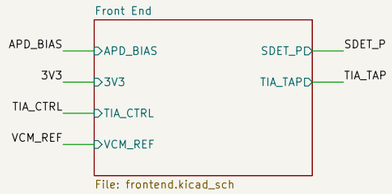
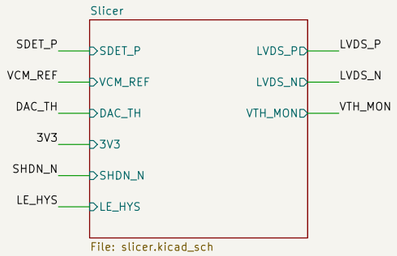
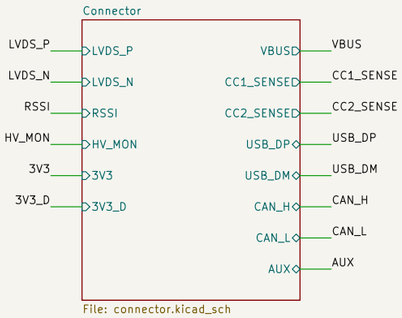
Root sheet. Signal flow: Power –> Front End –> Slicer/LVDS –> Connector. No AGC loop and no detector sheet (decision 12).
Monitor sheet buffers VCM_REF (1.65 V baseline) and the RSSI minimum detector from TIA_TAP; the Controller (ESP32-C6, decisions 11/12) sets
DAC_TH/DAC_HV/HV_DIS/SHDN_N/LE_HYS/TIA_CTRL, reads HV_MON/RSSI/VTH_MON, and talks CAN through the Connector; the single USB-C J50 on the Connector sheet carries VBUS, USB, LVDS, CAN and AUX (decision 14).
Status labels VERIFIED/PROPOSED/UNRESOLVED; no part values approved for fabrication.

Rails VBUS (Connector –> Power and Controller; the single 5 V input, decision 14), 3V3, 3V3_D are sheet pins; +5V is internal to Power (U5 TPS259631 eFuse OUT); GND is global. HV_ nets and APD_BIAS are net class HV (1.0 mm clearance).
LVDS_P is low for a logic one (post-amp non-inverting after an OPA857 whose OUT swings down with light); the FPGA inverts.

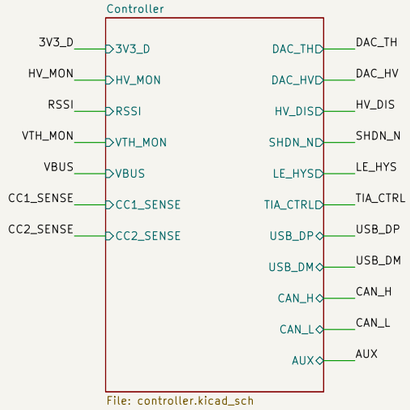














Global net GND enters here only to carry a PWR_FLAG for ERC. USB_DP, USB_DM, AUX, CAN_H, CAN_L run Connector <-> Controller; CC1_SENSE, CC2_SENSE Connector -> Controller.

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Sheet: /
File: receiver-module.kicad_sch

Title: Photodiode LVDS Receiver – Root

Size: A3 | Date: 2026-10-08 | **Rev: A0**
KiCad E.D.A. 10.0.5 | Id: 1/7

Power: VBUS in -> U5 eFuse -> +5V. U1 TPS79901 3V3, U4 TPS7A2033 3V3_D, U3 LT8365 APD bias (100–220 V) -> APD_BIAS.

U5 TPS259631DDAR eFuse (PROPOSED: TPS2596 SLVSET8A p.4, p.22):
IN = VBUS, OUT = +5V. R35 357k / R36 43.2k / R37 100k set UVLO and OVLO.
R38 1.21k sets ILM. C32 47 nF sets dVdt. C33 4.7 uF on VBUS.
FLT open-drain (TP4); R39 pull-up DNP. Not simulated or measured.

U5
VBUS
C33 4.7uF 25V
R35 357k 1%
R36 43.2k 1%
R37 100k 1%
C32 47nF X7R 16V
IN OUT EN/UVLO FLT OVLO ILM dVdt GND EP
TP3 IMON
+5V
3V3_D
TP4 FLT
R39 100k
R38 1.21k 1%
TPS259631DDAR

3V3_D: U4 TPS7A2033 3.3 V, 300 mA, own rail for ESP32-C6, CAN, DAC, ADC (about 60 mA est.). ESP32 radio MUST stay disabled (Wi-Fi TX peaks 251–382 mA). PROPOSED, estimate not measured.

U1
+5V
C1 10uF 10V
C2 100nF
C9 1uF
IN OUT EN GND FB
V3_FB
C10 4.7uF 10V
C11 4.7uF 10V
R1 17.8k 1%
R2 10.0k 1%
C12 1nF C0G
3V3
TPS79901DDCT

U4 TPS7A2033PDBV
+5V
C29 1uF 10V
IN OUT EN GND
C30 10uF 10V
C31 1uF 10V
3V3_D

Design intent (Rev A, revision 4; nothing here is simulated or measured; details and citations in docs/schematic/power.md)
PROPOSED +5V (U5 OUT) entry: C1 10 uF + C2 100 nF.
PROPOSED U1 TPS79901: VOUT = 1.193 V x (1 + R1/R2) = 3.316 V nominal (R1 17.8k, R2 10.0k). C (see note).
VERIFIED TPS79901 CFB is required for stability; 3 pF to 1 nF when R1||R2 = 250k; <5 pF for fast start (SBVS056L p.14, 8.2.2.6).
UNRESOLVED Here R1||R2 = 6.4k, not 250k: the datasheet range was not stated for this impedance; nF is a PROPOSED value.
PROPOSED 3V3 load estimate (rev 2): OPA857 26.8 mA max + OPA855 19.5 mA max (IQ 17.8 typ, 16 mA at VS = 5 V, OPA855 SBOS622D PDF p.7; Fig 6-27 p.12 shows about 16–17.5 mA at 3.3 V, sc is conservative) + TLV3605 17.5 mA max + OPA2320 ~2 mA = about 66 mA. U1 TPS79901 is rated mA. PROPOSED, not measured.
REMOVED (decision 12) U2 LM27762 +/-3V7, 5V_A bead/caps, PGOOD_VGA and TP1: the VCA821 VGA the OPA855 runs from 3V3.
PROPOSED U3 LT8365 values per docs/calculations/apd-bias-design.md sections 3.2 and 4: 500 kH 84.5k), L1 10 uH, boost to ~110 V + doubler.
VERIFIED SYNC/MODE: float = pulse-skip, GND = Burst, 100k to GND = Burst/SSFM (LT8365 Rev. A p BIAS tie to GND if unused: INTVCC 1 uF (p.7).
PROPOSED EN/UVLO 200k/200k from 5V_HVIN, always connected. Q3 DMN2056U-7 pulls EN/UVLO to OFF. Gate = HV_DIS, R33 100k pull-up to 3V3_D, R34 1k series: HV is OFF until the ESP32 drives HV_DIS low after verifying the DAC (decision 11). J1 HV FORCE ON (DNP) shorts the gate to GND for bench use without firmware.
PROPOSED U4 TPS7A2033 3V3_D: 300 mA, radio must stay disabled; load est. 60 mA (docs/research/module-controller.md). CIN 1 uF, COUT 10 uF + 1 uF (datasheet minimum 1 uF, r 200 uF: SBVS338H p.5).
PROPOSED BIAS: R28 OR to GND fitted; R29 OR to 5V_HVIN DNP. Fit exactly one of them. SYNC option R26/R27/TP2 are DNP.
PROPOSED D2 (GND to SW, from the p.30 250 V example, purpose not stated) is DNP. Q1/Q2 clamp uA. C8 and R19 form filter stage 1.
PROPOSED HV_MON = HV_OUT/101 (R20+R32 10M, R21 100k): 1.39 V at 140 V. Stage 2 filter (10k is on the front-end sheet.
Net names: HV_V1, HV_PUMP, HV_RAW, HV_OUT, HV_CLAMP, 5V_HVIN, EN, SYNC, BIAS, RT, SS, VC, FBX ar local labels; APD_BIAS = HV_FILT.
Hierarchical pins: in VBUS, DAC_HV, HV_DIS; out 3V3, 3V3_D, APD_BIAS, HV_MON.
PROPOSED (decision 14, revision 4) U5 TPS259631DDAR eFuse: IN pin 4 = VBUS (single 5 V input fro the Connector sheet). OUT pin 5 = +5V (internal rail for U1, U3 via L2, U4), GND pin 1 and exposed pad pin 9 = GND (TPS2596 SLVSET8A p.4). Fixed-clamp variants rejected; this variant gives x1 auto-retry (Device Comparison p.3). Abs max on IN 21 V (p.1, p.5).
PROPOSED UVLO/OVLO divider R35 357k / R36 43.2k / R37 100k (1 %): overvoltage cut-off 5.76-6. rising (below the 7.0 V abs max of the TPS79901, SBVS056L p.4), UVLO 4.06-4.32 V rising, 3.72-4.00 V falling (calculated with tolerances, not simulated).
PROPOSED ILM R38 1.21k 1 %: ILIM 0.68 / 0.76 / 0.84 A min/typ/max (TPS2596 Eq. 5 p.22). A real current limit, not the 0.5 A USB budget. TP3 IMON on ILM (about 0.79 V/A). C32 47 nF on dVdt: about 5.3 ms ramp, about 23 mA inrush into an assumed 25 uF (estimate, measure in G1).
PROPOSED FLT (pin 6) open-drain, TP4; R39 100k pull-up DNP to 3V3_D. UNRESOLVED: both rails are while the eFuse is cut off, so a pull-up reads 0 V and cannot separate fault from no power; FLT is not routed to the ESP32. FLT pin abs max not read.
PROPOSED C33 4.7 uF 25 V 0805 on VBUS at IN. C1/C2 stay on +5V (OUT). Removed: TPS2121, +5V PWR_SRC, old R39-R42, C34, the PR1 divider.
PROPOSED Budget about 0.19 A worst case from +5V (docs/research/usb-standalone-power.md sect 1.5, estimate) plus up to 35 mA on 3V3 for the LVDS fan-out U50 (LVDS104 p.5); radio must stay disabled. UNRESOLVED: whether a bus-powered hub holds a module to 100 mA.
PROPOSED ERC: U5 OUT (pin 5) is power_out on the +5V net, so +5V needs no PWR_FLAG; the root PWR_FLAG #FLG04 sits on VBUS. USB-alone firmware starts HV at the low-bias end in steps (PROPOSED).

+5V
L2 0.47uH 2.1A
C18 10uF 50V
C19 10uF 50V
C20 4.7uF 50V
PWR_FLAG #FLG02
5V_HVIN
L1 10uH 1.5A
D2 DFLS1150-7
D1 DFLS1150-7
R7 20R
HV_V1
C6 0.1uF 200V X7R
D3 DFLS1150-7
HV_PUMP
D4 DFLS1150-7
C7 0.1uF 200V X7R
HV_RAW
C5 0.1uF 200V X7R

U3
EN 1 EN/UVLO
5V_HVIN 3 VIN
BIAS 7 BIAS
SYNC 12 SYNC/MODE
RT 10 RT
SS 11 SS
SW1 16
SW2 14
INTVCC 5
FBX 9
VC 8
GND 17
C21 1uF 25V
LT8365EMSE

R30 47R
HV_OUT
C25 1nF 500V
R17 4.99M 1%
R18 4.99M 1%
Q1 MMBTA42
Q2 MMBT3904
R16 4.02k 1%
HV_CLAMP
R19 27k 1206 200V
C8 10nF 500V C0G
APD_BIAS

HV_OUT
R20 4.99M 1%
R32 4.99M 1%
R21 100k 1%
C28 10nF
HV_MON

5V_HVIN
R22 200k 1%
R23 200k 1%
EN
Q3 DMN2056U-7
R33 100k
3V3_D
R34 1k
HV_DIS
J1 HV FORCE ON
RT
R24 84.5k 1%
SS
C22 0.22uF
SYNC
R26 100k
R27 0R
SYNC_EXT TP2
BIAS
PWR_FLAG #FLG03
R28 0R
R29 0R

DAC_HV
R13 10k
C27 100nF
R14 442k 0.1%
FBX
C26 1pF C0G
R12 86.6k 1%
TH1 NTC 10k B3950
R8 2.05M 1%
R9 2.05M 1%
R10 2.05M 1%
R11 2.05M 1%
HV_OUT

VC
C24 1pF C0G
R25 84.5k 1%
C23 1nF C0G

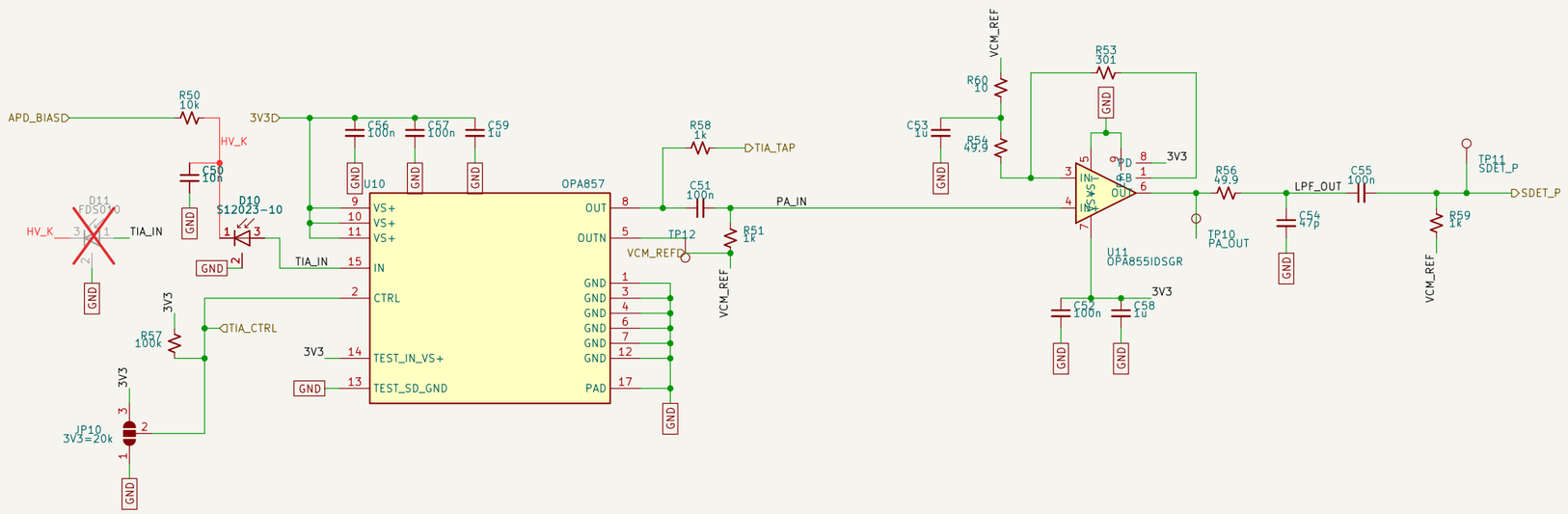
3V3
RV1 50k trim
R15 274k
FBX

HV WARNING
APD bias reaches 220 V. Stored energy ~2.3 mJ (docs/calculations/apd-bias-design.md section 4). C be hazardous.
Never probe HV_OUT, HV_RAW, HV_PUMP, HV_V1 or HV_CLAMP directly. Use HV_MON (HV_OUT/101) for observation.
Capacitors C5-C8, C25 and resistors R19, R30 carry >= 200 V ratings; keep 200 V creepage/clearanc on every HV_ net.
series-limited, with the HV netclass 1.0 mm clearance. Keep the APD_BIAS/FBX area away from TIA_IN.
HV is OFF by default (Q3 gate pulled up); J1 FORCE ON overrides the firmware disable: fit only on the bench with the DAC set. Discharge through the divider takes ~0.5 s: wait before touching.
References: LT8365 Rev. A datasheet p.30 (250 V example), DC2838A demo manual Rev. 0.

Aharoni Lab, UCLA
Sheet: /Power/
File: power.kicad_sch
Title: Power — Photodiode LVDS Receiver
Size: A3 | Date: 2026-10-08 | **Rev: A0**
KiCad E.D.A. 10.0.5 | Id: 2/7

Front end: APD -> OPA857 TIA -> AC coupling -> OPA855 x7 -> low-pass -> AC coupling -> SDET_P (baseline VCM_REF)

1 APD + HV filter 2 OPA857 TIA, 3.3 V 3 AC coupling + RSSI tap 4 OPA855, G = 1 + 301/49.9 = 7 5 low-pass 68 MHz 6 AC coupling



CTRL: R57 gives 20k; JP10 B-C forces 20k, A-C forces 5k

Design intent (revision 2, decision 12; nothing here is simulated or measured)

PROPOSED Chain: S12023-10 APD, OPA857 TIA (20k/5k), AC coupling C51/R51, OPA855 non-inverting gain 7, 68 MHz low-pass, AC coupling C55/R59 to SDET_P. DC level of the whole post-amp path is VCM_REF (1.65 V from the Monitor sheet), so the output idles at the comparator baseline.

PROPOSED Stage 2 HV filter R50 10k (1206, 200 V) + C50 10n (1210, 500 V C0G) at the cathode; stage 1 is on the Power sheet. HV_K is up to 220 V: keep-out to TIA_IN (200 V creepage/clearance), no HV test points.

VERIFIED OPA857 (SBOS630D): bias APD cathode to positive supply, anode to IN, so the output swings DOWN below OUTN = 1.83 V (p.19, 8.2.1.2 step 2; p.15, 7.3.1). Test_SD (13) GND, Test_IN (14) 3V3 (p.4 pin table). CTRL 1 = 20k, 0 = 5k (p.6).

VERIFIED OPA855 (SBOS622D): FB (pin 1) is internally the amplifier output; RF goes FB to IN- on the same side of the package. OUT drives the load, NC pin 2 stays open (p.17, 8.3.2, Fig. 8-4). PD high = on, enable threshold 1.5 V typ / 1.8 V max above VS-; PD bias 70 uA typ, 140 uA max (p.7), so PD goes straight to 3V3 (a 10k would drop 0.7-1.4 V). Thermal pad to VS- (p.20, 8.4.1; p.29).

VERIFIED OPA855 input common-mode 1.3 V max low to 2.7 V min high at VS+ = 3.3 V (p.6): VCM_REF 1.65 V is inside. Overdrive recovery 3 ns (p.6). Datasheet gain-7 test values are RF 453 / RG about 75 (Fig. 7-3, p.15); no RF/RG table exists, so 301/49.9 (G = 7.03) is PROPOSED: lower resistances load the output less than the test values on a 3.3 V part. UNRESOLVED: stability with RF 301 not checked.

PROPOSED RG returns to PA_REF = VCM_REF through R60 10 ohm with C53 1u to GND, so gain is referenced to the baseline and the 1u cap does not load the Monitor sheet's VCM_REF buffer. Input return R51 1k to VCM_REF (corner 1.6 kHz, 100 us). OPA855 IB -12 uA typ (p.6) into 1k gives about +80 mV output offset (derived), which moves the idle point to about 1.73 V, close to the 1.75 V mid-swing of the 1.15 to 2.35 V output limits (p.7).

PROPOSED Clip level (derived): 1.15 Vpp / (7 x 18.2k) = 9 uA at 20k, 36.5 uA at 5k; OPA857 saturates at 60 uA / 240 uA (p.15, 7.3.1). Output is AC coupled after the clipping amplifier so clipping is symmetric about the baseline.

PROPOSED Low-pass R56 49.9 + C54 47p = 68 MHz (research 60-80 MHz). SDET_P: C55 100n, R59 1k to VCM_REF (matched to the 1k on the Slicer's IN- side). TP10 PA_OUT, TP11 SDET_P, TP12 TIA_REF. TIA_TAP = TIA_OUT through R58 1k to the RSSI detector.

UNRESOLVED OPA857 stability and bandwidth at 6 pF source capacitance (datasheet characterises 1.5 pF, p.6). OPA855 stability at G = 7 into the 49.9 ohm / 47 pF load not simulated. Received power not measured.

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Sheet: /Front End/
File: frontend.kicad_sch

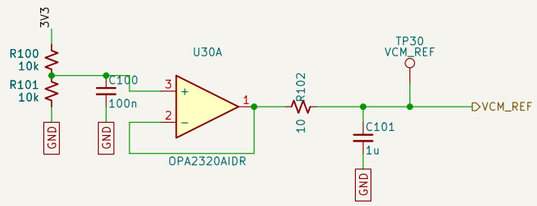
Title: Front End - Photodiode LVDS Receiver

Size: A3	Date: 2026-10-08	**Rev: A0**
KiCad E.D.A. 10.0.5		Id: 3/7

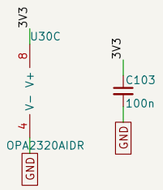
Monitor: VCM_REF baseline reference (U30A) and RSSI peak detector (D30, U30B). Left: VCM_REF; right: RSSI.

3V3D 3V3

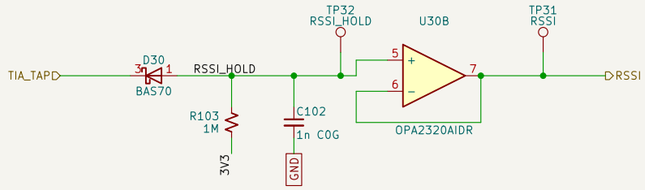
1. VCM_REF



R100/R101 10k 1 % (or better): 3V3/2 = 1.65 V nominal.
R102 10 ohm isolates C101 1 uF; VCM_REF is taken AFTER R102.



2. RSSI (negative-peak detector)



D30 BAS70: K on TIA_TAP, A on RSSI_HOLD. OPA857 OUT swings DOWN from OUTN (1.83 V) on photocurrent, so the detector holds the NEGATIVE peak; RSSI falls as the light rises. 1 M (to 3V3) x 1 nF = tau 1 ms.

DESIGN INTENT (Monitor sheet, revision 2, decision 12). Status labels per CLAUDE.md. Nothing here is measured or simulated.

WHAT THIS SHEET IS FOR
VCM_REF is the common baseline of every AC-coupled stage downstream of the OPA857: the return of the post-amp input coupling (Front End), the OPA855 gain reference, and the comparator input returns (Slicer). It sits at 3V3/2 = 1.65 V so the TLV3605 inputs are mid-range. With 8b/10b the eye centre is the AC-coupled baseline, so no tracking threshold is needed (docs/research/post-amp-slicer.md).
RSSI is a slow, RELATIVE received-light indicator for the ESP32 (read by the ADS1115) to trim APD bias (DAC_HV) and the OPA857 20k/5k gain. It is not calibrated: the BAS70 forward drop and the 1 k tap resistor on the Front End (R58) shift it; the ESP32 calibrates against the FPGA's 8b/10b validity.

VCM_REF (PROPOSED)
U30A OPA2320 unity buffer of a 10k/10k divider from 3V3 with C100 100 nF at the midpoint. U30A drives C101 1 uF through R102 10 ohm and VCM_REF is taken after R102.
Reason: the unity-gain OPA320 is stated stable with a pure capacitive load up to about 1 nF, and a 10 to 20 ohm series resistor is the datasheet's technique for larger loads (SBOS513F, 7.3.8, PDF p.19, Figure 36); ESR of caps above 1 uF can also help. A 1 uF load is above the datasheet's own illustrated cases, so the margin is UNRESOLVED until the VCM_REF step response is measured. The loads are a few 1 k returns to VCM_REF (microamps in use, 1.65 mA each only if one were shorted to ground), so the 10 ohm drop is negligible.

RSSI (PROPOSED)
OPA857 OUT swings down below OUTN (1.83 V) for photocurrent into IN (SBOS630D, 7.3.1 PDF p.15; 8.1 PDF p.17; 8.2.1.2 step 2 PDF p.18), so D30 (cathode on TIA_TAP, anode on RSSI_HOLD) captures the negative peak. R103 1 M returns RSSI_HOLD to 3V3 and C102 1 nF C0G gives tau 1 ms. At idle R103 passes about (3.3 - 1.83) V / 1 M = 1.5 uA through D30 and the 1 k tap into the TIA output: about 1.5 mV, negligible. RSSI_HOLD therefore idles at about OUTN + Vf (roughly 1.9 to 2.0 V; Vf at 1.5 uA not looked up) and falls as the light rises. TIA_TAP is DC-coupled, so the pull-up cannot saturate anything. U30B buffers RSSI_HOLD to RSSI (unity gain).
PROPOSED No dead zone: with R103 to VCM_REF the diode would stay off for the first 0.3 to 0.5 V of swing (TIA idles at 1.83 V, RSSI_HOLD at 1.65 V). The 3V3 return keeps D30 slightly conducting so RSSI tracks the TIA minimum from the first microamp. Cost: the idle level depends on Vf (relative indicator, the ESP32 calibrates). Not simulated or measured.

REMOVED (decision 12)
VMID, VSPAN, VCM_DET, VTARGET, VG_HI/VG_LO, VLOS_TH references, the AGC integrator, TMUX1112 freeze switches, TLV3201 LOS comparator and the VG/VTH selectors: the VCA821 VGA and analog AGC are gone, so nothing needs them. Range control is a firmware trim from RSSI.

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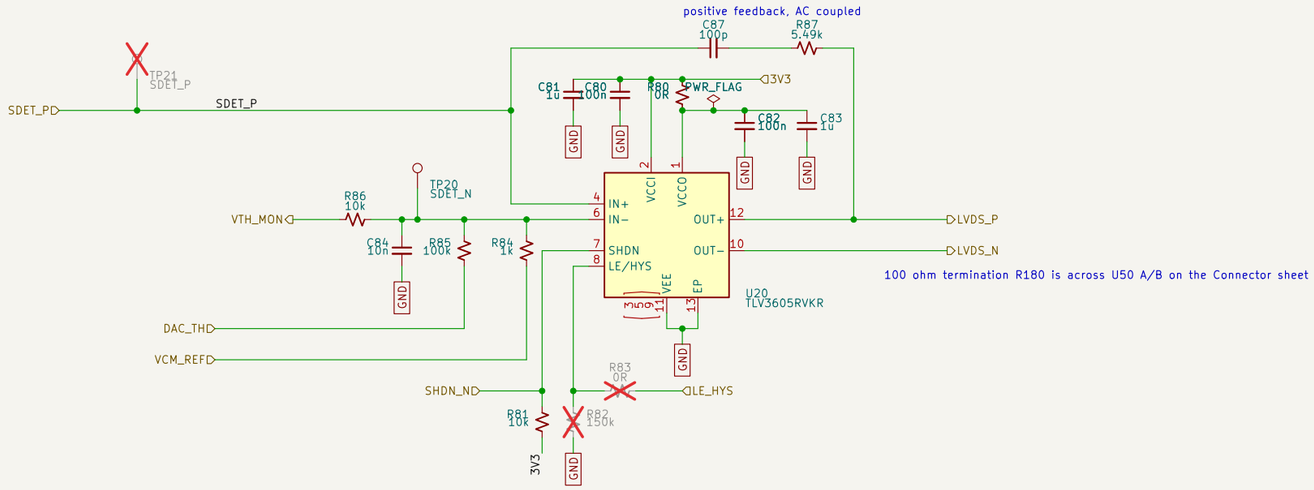
Sheet: /Monitor/
File: control.kicad_sch
Title: Monitor - Photodiode LVDS Receiver
Size: A3 | Date: 2026-10-08 | **Rev: A0**
KiCad E.D.A. 10.0.5 | Id: 5/7

Slicer: SDET_P is compared with the VCM_REF baseline (SDET_N, trimmed by DAC_TH) in a TLV3605 with 3 mV external hysteresis; LVDS out.

1 inputs: SDET_P, threshold network SDET_N

2 TLV3605 comparator

3 LVDS out



100 ohm termination R180 is across U50 A/B on the Connector sheet

Design intent (revision 2, decision 12; nothing here is simulated or measured)

PROPOSED The signal arrives already AC coupled and referenced to VCM_REF (Front End sheet). With 8b/10b the eye centre is that baseline, so IN− is the same baseline plus a small trim: no peak tracking. Light on makes SDET_P go BELOW the baseline (OPA857 output swings down, SBOS630D p.19), so LVDS_P is LOW for a one: the FPGA must invert, or LVDS_P/N be swapped at the FPGA.

PROPOSED Threshold SDET_N: R84 1k from VCM_REF (matches the 1k return R59 on IN+, so bias currents cancel: IB −1 uA typ / +−5 uA max, IOS +−1 uA, SNOSDA2F p.7; a single 1k would give 1 mV typ / 5 mV max, 10k would give 10 / 50 mV). DAC_TH through R85 100k: trim = (DAC − VCM_REF) x 1k/101k, so a 0 to 3.3 V DAC with VCM_REF 1.65 V moves the threshold +−16.3 mV (8 uV per 12−bit LSB). C84 10n to GND keeps IN− quiet (corner 16 kHz). VTH_MON = SDET_N through R86 10k for the ADC.

PROPOSED External hysteresis (research 2−4 mV: the internal LE/HYS pin cannot go below about 8 mV, SNOSDA2F p.8, Fig. 7−4 p.16): positive feedback from OUT+ to IN+ through R87 5.49k in series with C87 100p. OUT+ swings 350 mVpp (VOD 350 mV typ, VOCM 1.2 V, p.7). Source seen at IN+ at signal speed = 49.9 ohm (R56) || 1k (R59) = 47.5 ohm. Window = 350 mV x 47.5/(5490 + 47.5) = 3.0 mV. C87 blocks DC: a bare resistor would shift IN+ by (1.2 − 1.65) x 1k/(1k + 5.49k) = −69 mV. High−pass corner 1/(2 pi x 5.54k x 100p) = 287 kHz, tau 554 ns, so a 100 ns run (5 UI) keeps 83 % of the window. The spec text of about 100k with a 1k source only holds for slow (DC) states: at 50 MHz edges it gives 0.17 mV. UNRESOLVED: effect on jitter not simulated.

PROPOSED LE/HYS (pin 8) left open = internal hysteresis 0 mV and latch off (p.8). R82 150k to GND (30 mV, DNP) and R83 0R to LE_HYS (DNP) are footprints only. SHDN (pin 7): R81 10k pull−up, SHDN_N active low (VIH 1.5 V, VIL 0.4 V, p.8); shutdown is not a guaranteed zero output (standing correction 11).

VERIFIED VCCI = VCCO = 3V3 (2.4 to 5.5 V, p.19); 100 nF 0402 + 1 uF at each of VCCI and VCCO, VCCO through R80 0R (bead option). VEE and EP to GND (datasheet gives no pad requirement beyond a ground plane). PWR_FLAG on the VCCO net because R80 is passive.

PROPOSED LVDS: OUT+ = LVDS_P, OUT− = LVDS_N; 100 ohm termination R180 across the A/B inputs of the fan−out buffer U50 on the Connector sheet, a few cm away (revision 4; SN65LVDS104 has no on−chip termination, LVDS104 p.16 Fig. 25; the ECP5 end is terminated with DIFFRESISTOR, docs/research/ulx3s−lvds−input.md). TP20 SDET_N, TP21 SDET_P (DNP, the probe loads the input).

UNRESOLVED Comparator input noise and delay below 10 mV overdrive are not in the datasheet (docs/research/post−amp−slicer.md section 1); IB at VCM = 1.65 V is not specified.

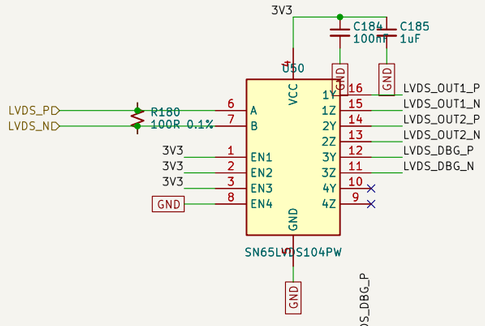
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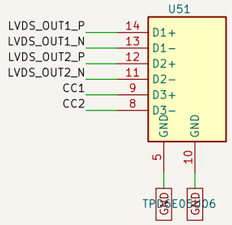
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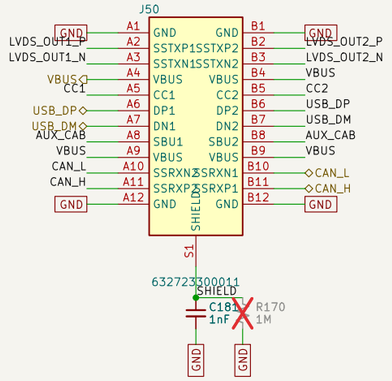
Title: Slicer / LVDS − Photodiode LVDS Receiver

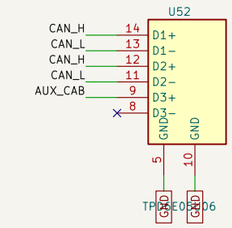
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KiCad E.D.A. 10.0.5		Id: 6/7

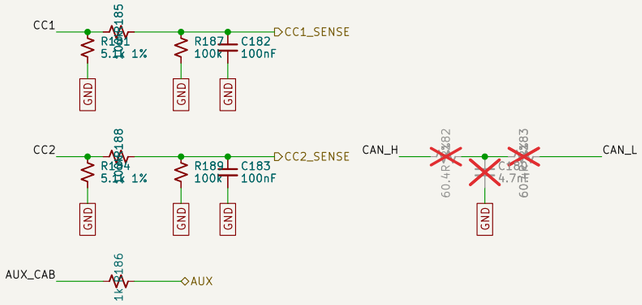
Connector: J50 USB-C (Wuerth 632723300011, 24-pin), U50 LVDS fan-out, U51/U52 ESD, J51 scope header (revision 4, decision 14).

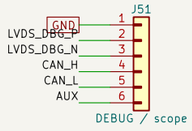


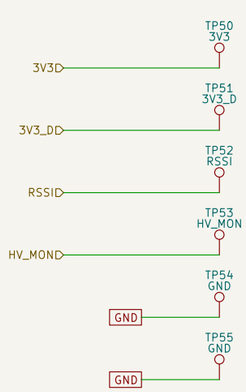












Design intent (Rev A revision 4, 2026-10-09, decision 14: PROPOSED unless labelled; nothing here is simulated or measured)
ONE CONNECTOR: J50 carries 5 V (VBUS pins A4/A9/B4/B9 tied, hierarchical output VBUS to the Power eFuse and the Controller), USB 2.0 D+/D- (A6/B6 tied = USB_DP, A7/B7 tied = USB_DM), the 50 Mbps LVDS pair on both SuperSpeed TX pairs through the fan-out U50 (1Y/1Z -> A2/A3, 2Y/2Z -> B2/B3), CAN on both RX pairs (CAN_H = B11 and A11, CAN_L = B10 and A10) and the SBU pair tied as AUX (A8/B8 -> AUX_CAB -> R186 1k -> AUX). Plug orientation does not matter: P to SSTXp and N to SSTXn on both pairs (USB Type-C R2.5 Table 3-10 p.83).
Rd/CC: R181 and R184 5.1 k 1 % on CC1 and CC2 (Type-C R2.0 Table 4-25 p.236). Taps 100k/100k with 100 nF give CC1_SENSE and CC2_SENSE for the ESP32 ADC; 100k/100k keeps the pin below 3.3 V even with 5.5 V VCONN on the unused CC (TC R2.5 p.155). The 200 k tap lowers Rd by about 2.5 %: fold into firmware thresholds.
U50 SN65LVDS104PW (LVDS104 SLLS396G p.3-5, p.16 Fig. 25): no on-chip termination, so R180 100 ohm 0.1 % across A/B is FITTED; enables active high (section 9.4 Table 2), EN1-EN3 to 3V3, EN4 to GND (output 4 disabled, 4Y/4Z no_connect), output 3 feeds J51 with R177 100 ohm DNP across it (UNRESOLVED whether an unloaded output 3 is acceptable), 23 mA typ, 35 mA max (p.5). VOD 250/350/450 mV (TLV3605 p.7) against the 100 mV buffer threshold: 150 mV margin at minimum VOD (arithmetic). Fail-safe: static high with no TLV3605 power (p.15-16). Do NOT add a second 100 ohm at the FPGA without DIFFRESISTOR consideration: the ECP5 input is terminated on-die (DIFFRESISTOR 100, docs/research/ulx3s-lvds-input.md); each of the two paths sees one load.
ESD: U51 (SSTX1, SSTX2 pairs, CC1, CC2) and U52 (CAN on both RX pairs, SBU AUX_CAB, one spare channel) TPD6E05U06RVZ (TI SLVSBO7O Table 5.4 p.7: 0.47 pF, VRWM 5.5 V): place within a few mm of J50. D+/D- clamp is U61 on the Controller sheet.
CAN: no termination on the module (SLLA270 p.4, p.8): R182/R183/C180 are DNP for a single-module bench; never fit them on a module in a star. 125 kbit/s PROPOSED. Two bused pairs give about 45 ohm in parallel, acceptable at the slowed edges.
Shield: S1 to GND through C181 1 nF with R170 1 M (DNP). Layout rules (not schematic): Dp/Dn ties within 3.5 mm of J50 (TC Table 3-4 note 1 p.77).
UNRESOLVED: top- or mid-mount J50; SBU exposure to a laptop alternate mode; unidirectional ESD clamp near -0.7 V versus cable GND drop on CAN; the 3.3 V-only ESP32 GPIO behind R186 1k.

J50 pin map (WE 632723300011 p.3, TC Table 3-4 p.76: A5/B5 are labelled CC/VCONN by Wuerth, wired as CC1/CC2: PROPOSED reading)
A1 GND A2 SSTXP1=LVDS_OUT1_P A3 SSTXN1=LVDS_OUT1_N A4 VBUS A5 CC1 A6 DP1=USB_DP A7 DN1=USB_DM A8 SBU1=AUX_CAB A9 VBUS A10 SSRXN2=CAN_L A11 SSRXP2=CAN_H A12 GND
B1 GND B2 SSTXP2=LVDS_OUT2_P B3 SSTXN2=LVDS_OUT2_N B4 VBUS B5 CC2 B6 DP2=USB_DP B7 DN2=USB_DM B8 SBU2=AUX_CAB B9 VBUS B10 SSRXN1=CAN_L B11 SSRXP1=CAN_H B12 GND
J51 DEBUG (2.54 mm 1x6): 1 GND, 2 LVDS_DBG_P (U50 3Y), 3 LVDS_DBG_N (U50 3Z), 4 CAN_H, 5 CAN_L, 6 AUX.
Test points TP50-TP55: 3V3, 3V3_D, RSSI, HV_MON, GND, GND. DNP: R170, R177, R182, R183, C180.
Cable rule: passive full-featured USB 3.2 Gen 1, 3 A, e-marked, <= 2 m certified; 3 m only after a bench check (TC Table 3-1 p.44, 3.4.1.1 p.83, 4.4.1 p.151). Labelling: USB-C: USB 2.0 and 5 V only on a host; SuperSpeed pairs carry proprietary LVDS/CAN, not USB 3; the module is not USB 3 compliant.
HV rule: no bare APD_BIAS pad on this sheet; HV_MON (about Vout/101) is the only HV observation point.
ULX3S target pairs (PROPOSED; ULX3S schematics v3.1.7 p.2, Lattice FPGA-TN-02032 v1.4 p.14, p.26): first choice gp/gn[8..13] (bank 7), second gp/gn[14..21]; not [0..7] or [22..27]. Constrain gp only; do not guess polarity.

Aharoni Lab, UCLA
Sheet: /Connector/
File: connector.kicad_sch
Title: Connector — Photodiode LVDS Receiver
Size: A3 | Date: 2026-10-08 | Rev: A0
KiCad E.D.A. 10.0.5 | Id: 7/7

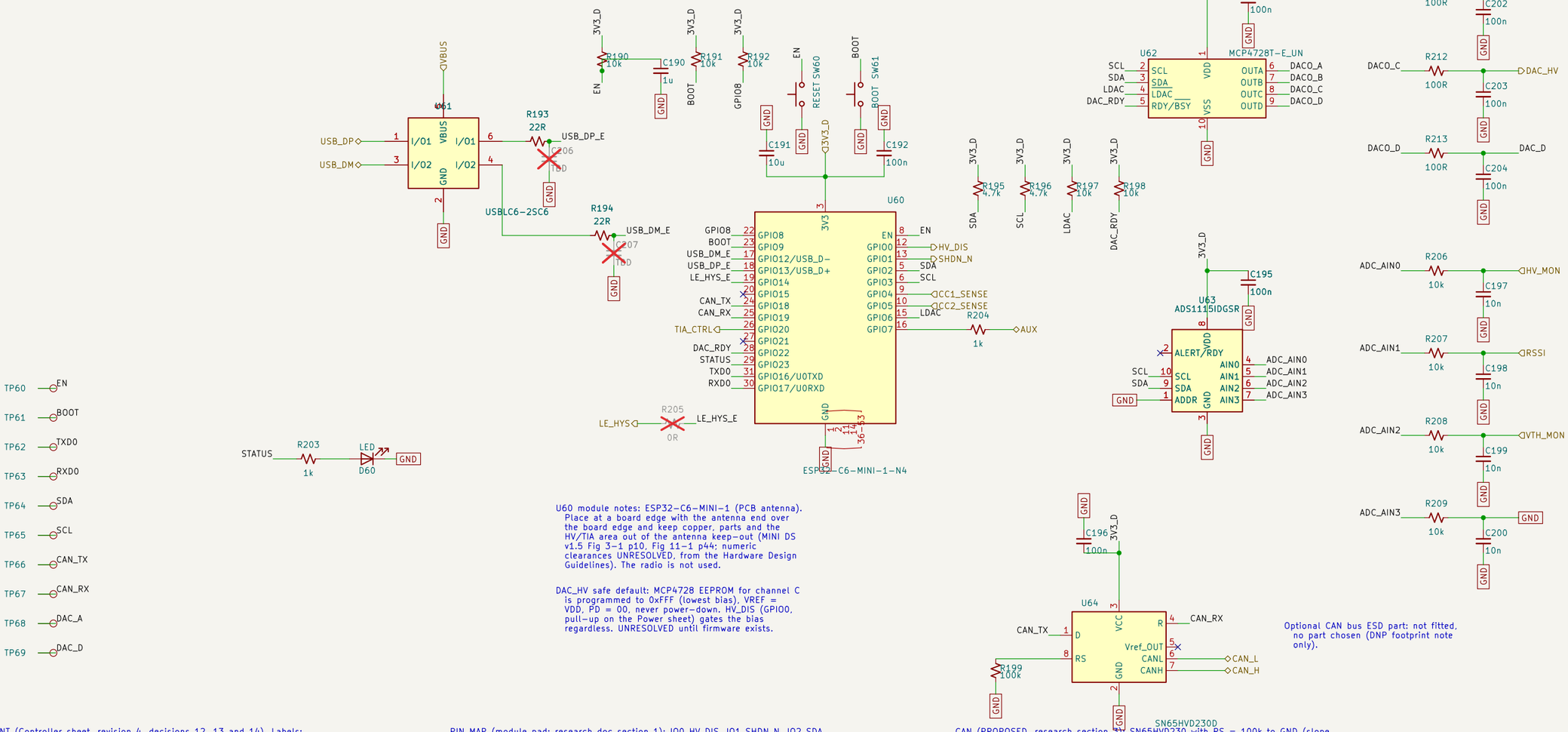
USB D+/D- (PROPOSED, decision 14): USB_DP/USB_DM arrive from J50 on the Connector sheet (hierarchical, bidirectional). R193/R194 22R (ESP32-C6 Hardware Design Guidelines p.18 section 1.3.13: reserve 22/33 ohm and capacitors to ground: MINI-1 datasheet v1.5 Fig. 9-1 p.40 leaves them TBD), with DNP C206/C207 (0402, value TBD) from USB_DP_E/USB_DM_E to GND on the module side. UNRESOLVED: effect of 22 ohm with USBLC6 and auto-reset through them, untested. VBUS surge protection is the eFuse U5 on the Power sheet (21 V abs max, TPS2596 p.5), not the USBLC6; USBLC6-2SC6 VBUS-pin clamp behaviour UNRESOLVED (ST datasheet not read). Place U61 near J50 at layout.

BUTTONS (PROPOSED): SW60 RESET (EN to GND) and SW61 BOOT (IO9 to GND). C&K PTS810 SJM 250; TP60/TP61 in parallel. R190 10k + C190 1u is the debounce and the recommended EN RC (MINI-1 datasheet v1.5 p.40, HDG p.10). Download boot = GPIO9 0 with GPIO8 1 at EN rising (MINI-1 datasheet Table 4-3 p.13), VERIFIED. Fitted even though RTS/DTR auto-reset exists (TRM v1.2 p.1030 Table 32.3-2): recovery for empty flash or a disabled USB PHY (HDG p.30).

CC SENSE (PROPOSED, decision 14): CC1_SENSE (IO4, ADC1_CH4) and CC2_SENSE (IO5, ADC1_CH5) arrive from the Connector sheet already divided 100k/100k with 100 nF; only the 0-1.9 V ATTEN2 range is read (ESP32-C6 datasheet v1.5 Table 5-6 p.66, +-23 mV after calibration), so Rp classification is thin (firmware: three agreeing reads). IO4/IO5 are strapping pins (MTMS/MTDI) that only select SDIO sampling edges (MINI-1 datasheet Table 4-4 p.14); SDIO unused, PROPOSED harmless. UNRESOLVED: finer ADC attenuation range; clamp-injection current for a fault on the CC taps (input high max VDD + 0.3 V, Table 5-4 p.65). VBUS_SENSE and PWR_SRC are removed.

FIRMWARE STATES (PROPOSED, firmware/esp32/README.md): cc_class, cc_orient and usb_host are logged hints; the definitive system indicator is LINK_ALL from the FPGA master. The ESP32 must not clear HV_DIS on the basis of CC levels alone. UNRESOLVED: firmware policy for the D+ pull-up backdrive into an unpowered host (TRM p.1042); self-test cannot prove the comparator or LVDS output without an FPGA or optical source; ground offset between laptop and FPGA supply not analysed; EN-side ESD diode; eFuse dVdt ramp (about 5.3 ms estimated) versus the 10 ms EN RC.

Strapping (MINI DS Table 4-1/4-3 p12-13): GPIO9 high = SPI boot; GPIO9 low with GPIO8 high = download boot. EN: 10k + 1u RC power-up delay (p40). GPIO15 left open (default eFuses ignore it).



U60 module notes: ESP32-C6-MINI-1 (PCB antenna). Place at a board edge with the antenna end over the board edge and keep copper, parts and the HV/TIA area out of the antenna keep-out (MINI DS v1.5 Fig 3-1 p10, Fig 11-1 p44; numeric clearances UNRESOLVED, from the Hardware Design Guidelines). The radio is not used.

DAC_HV safe default: MCP4728 EEPROM for channel C is programmed to 0xFFF (lowest bias), VREF = VDD, PD = 00, never power-down. HV_DIS (GPIO0, pull-up on the Power sheet) gates the bias regardless. UNRESOLVED until firmware exists.

Optional CAN bus ESD part: not fitted, no part chosen (DNP footprint note only).

DESIGN INTENT (Controller sheet, revision 4, decisions 12, 13 and 14). Labels: VERIFIED = datasheet cited, PROPOSED = design intent, UNRESOLVED = open. Nothing here is measured or simulated.
ROLE (PROPOSED): on-module ESP32-C6 does the slow trims and housekeeping, never the data path. (1) Range trim: DAC_HV (MCP4728 ch C) moves the APD bias toward the Q(M) optimum, and TIA_CTRL (IO20) schedules the OPA857 gain 20k / 5k from RSSI (ADS1115 AIN1) and FPGA link validity (received over CAN). (2) Threshold trim: DAC_TH (ch B) sets the Slicer threshold node; VTH_MON (AIN2) reads it back. (3) HV gating: HV_DIS (IO0) enables the APD bias supply; HV_MON (AIN0) reads the bias. (4) Module identity and status over CAN (CAN_H/CAN_L, 125 kbit/s), plus SHDN_N and LE_HYS to the Slicer. DAC_A (ch A) and DAC_D are spare test pads (TP68, TP69); ADS1115 AIN3 is tied to GND through its RC (spare).
UNPROGRAMMED ESP32 (PROPOSED): the module decodes. TIA_CTRL floats, so the Front End pull-up gives 20 k; SHDN_N pull-up keeps the comparator on; the Slicer threshold sits at VCM_REF through its resistor network. BUT HV_DIS is pulled up on the Power sheet, so an unprogrammed module has NO APD bias and no gain. Bench jumper J1 (HV FORCE ON, Power sheet) forces HV on. DAC_HV default (EEPROM) is the lowest bias.

PIN MAP (module pad; research doc section 1): IO0 HV_DIS, IO1 SHDN_N, IO2 SDA, IO3 SCL, IO6 LDAC, IO7 AUX via R204 1k, IO14 LE_HYS via R205 (0R, DNP), IO20 TIA_CTRL (direct, pull-up on the Front End), IO21 no_connect (AUX2 removed), IO8 strap pull-up, IO9 BOOT, IO12/IO13 USB D-/D+, IO16/IO17 TXD0/RXD0 pads, IO18 CAN_TX, IO19 CAN_RX, IO22 DAC_RDY, IO23 status LED, IO4 CC1_SENSE and IO5 CC2_SENSE (hierarchical inputs from the Connector). IO15 and IO21 are no_connect.
CHANGE FROM EARLIER SPEC: IO10 and IO11 do not exist on the MINI-1 (MINI DS Table 3-1 p10-11; 22 GPIOs), so LDAC is IO6 and AUX is IO7 (JTAG MTCK/MTDO pins; USB-Serial-JTAG is the debug path). IO15 is a strapping pin, so TIA_CTRL uses IO20. PROPOSED.
STRAPPING (VERIFIED, MINI DS Table 4-1..4-3 p12-13): GPIO9 weak pull-up, high = SPI boot; download boot needs GPIO8 = 1 and GPIO9 = 0 (BOOT pad to GND, then EN pad). GPIO8 has R192 10k to 3V3_D. Do not drive IO8/IO9 at reset. IO0, IO1, IO14 are Hi-Z at reset (C6 DS Table 2-2 p17-18), so the pull resistors on the other sheets fix the safe state. IO18-IO23 have weak pull-ups at reset (CAN_TX idles recessive; TIA_CTRL reads high = 20 k even before the Front End pull-up).
USB (PROPOSED, decision 14): the receptacle is J50 on the Connector sheet; USB_DP/USB_DM, VBUS (U61 pin 5 only) and CC1_SENSE/CC2_SENSE arrive as hierarchical inputs. D+/D- through R193/R194 22R with DNP C206/C207 to GND (details in the notes at the top). UART fallback on TP62/TP63.

CAN (PROPOSED, research section 3): SN65HVD230 with RS = 100k to GND (slope control, 600/800/1200 ns rise, SLOS346O p8; stub rule SLLA270 p9 allows at least 20 m, PROPOSED arithmetic; revision 4). 125 kbit/s. NO termination on the module; one 120 ohm (or 2 x 60 ohm + 4.7 nF split, SLLA270 p8) at the FPGA-end junction only. Vref_OUT unconnected.
I2C (VERIFIED, MCP4728 DS22187E p21-24; ADS1115 SBAS444E p20): private bus, MCP4728 at 0x60, ADS1115 ADDR = GND at 0x48. MCP4728 factory default is 0 V on all channels (not Hi-Z). ADS1115 inputs must stay within GND-0.3 .. VDD+0.3 V; the three sources are <= 3.3 V nodes. LDAC is held high by R197 so a high LDAC keeps the EEPROM values on the outputs.
UNRESOLVED: USB series resistor effect and auto-reset over USB; DAC_TH safe direction; verification of the DAC_HV default after EEPROM programming (General Call Read); firmware range-trim and TIA_CTRL scheduling algorithm; measured 3V3_D current; antenna keep-out numbers; -1 versus -1U module; VBUS back-feed against the 5 V cable rail.

Aharoni Lab, UCLA
Sheet: /Controller/
File: controller.kicad_sch
Title: Controller (ESP32-C6, DAC, ADC, CAN)
Size: A3 | Date: 2026-10-08 | **Rev: A1**
KiCad E.D.A. 10.0.5 | Id: 8/7